



华南理工大学

South China University of Technology

《统计学习与数据科学课程设计》课程论文

(2025-2026 学年第 2 学期)

**论文题目： Enzyme Kinetic Parameter Prediction
Based on Statistical Learning and Multimodal
Representation Learning**

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2、论文要求自己动手撰写，如发现论文是从网上下载的，或者是抄袭剽窃别人文章的，按作弊处理，本门课程考核成绩计 0 分。

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本科生课程论文

Title: Enzyme Kinetic Parameter Prediction Based on Statistical Learning and Multimodal Representation Learning

Student Name: YiZhang Mo

Abstract: Enzyme kinetic parameters, including turnover number (k_{cat}), Michaelis constant (K_m), and inhibition constant (K_i), are key quantitative indicators for understanding catalytic efficiency, substrate affinity, and inhibitor effects. Accurate prediction of these parameters is of great importance in enzyme engineering, metabolic pathway optimization, drug discovery, and industrial biotechnology.

Recent advances in machine learning, especially protein language models and multimodal learning frameworks, have significantly improved the prediction of biochemical properties from sequence and molecular data. Representative studies such as CatPred and UniKP demonstrate that integrating enzyme sequences, substrate structures, and structural context can outperform traditional methods.

This project proposes a progressive framework that combines classical statistical learning baselines with modern multimodal representation learning ideas. In the first stage, interpretable regression models will be established using engineered biological features. In the second stage, pretrained embeddings and advanced models may be explored. The project aims to build a reproducible, explainable, and extensible enzyme kinetic prediction pipeline suitable for scientific applications.

Keywords: Enzyme Kinetics; Machine Learning; Statistical Learning; Multimodal Learning; Regression Analysis; Bioinformatics; Protein Language Model; CatPred

1. Introduction

1.1 Research background

Enzymes are biological catalysts that govern nearly all metabolic reactions in living systems.

Their catalytic behavior is quantitatively described by enzyme kinetic parameters.

The most important parameters include:

k_{cat} : turnover number

K_m : Michaelis constant

K_i : inhibition constant

These quantities are fundamental in: Rational enzyme engineering, Drug screening,

Metabolic network simulation, Industrial catalyst optimization, Synthetic biology.

Traditional acquisition of such parameters requires wet-lab biochemical assays, which are expensive, slow, and experimentally constrained.

Therefore, computational prediction has become an important AI-for-Science task.

1.2 Literature Motivation

Recent benchmark studies show promising progress:

CatPred (2025): comprehensive deep learning framework for kinetic parameter prediction.

UniKP (2023): unified prediction framework for enzyme kinetics.

ERBA (2026): multimodal protein language model integrating substrate recognition and conformational adaptation.

These studies indicate that combining statistical learning with biological representation learning is a practical direction.

2. Research Objectives

This project aims to:

1. Build reproducible baseline regression models for enzyme kinetics prediction.
2. Predict $\log_{10}(k_{cat})$ as the initial endpoint.
3. Compare interpretable statistical models and ensemble models.
4. Explore multimodal extensions combining sequence and substrate features.
5. Analyze biological factors influencing prediction performance.
6. Produce publication-style figures and scientific reporting.

2. Problem Formulation

Given an enzyme-substrate pair:

Enzyme sequence S_e

Substrate representation S_m

Predict:

$$\hat{y} = \log_{10}(k_{cat})$$

Optional multi-target extension:

$$\hat{Y} = [\log_{10}(k_{cat}), \log_{10}(K_m), \log_{10}(K_i)]$$

This is a supervised nonlinear regression task.

4. Proposed Research Framework

Stage A: Classical Statistical Learning Baselines

Input handcrafted biological features into:

1. Linear Regression
2. Ridge Regression
3. Lasso Regression
4. Random Forest Regressor

5.XGBoost Regressor

Stage B: Representation Learning Extension

Use pretrained embeddings:

Protein sequence embeddings (ESM / ProtT5)

Molecular fingerprints / RDKit descriptors

Then train downstream regressors.

Stage C: Multimodal Fusion (Optional)

Fuse sequence and substrate embeddings using:

Concatenation fusion

MLP fusion network

Attention-based fusion

5. Technical Roadmap

Raw Data → Data Cleaning → Feature Engineering → Baseline Models → Advanced
Embedding Models → Model Comparison → Interpretability Analysis → Final Report

6. Dataset Plan

Data Acquisition Strategy

The project will first use the official CatPred benchmark dataset as the primary source. Data files will be downloaded from the official OSTI repository and stored in a structured local data warehouse.

Additional public databases such as BRENDA, SABIO-RK, and UniProt may be used for supplementary annotation and feature enrichment if needed.

All downloaded raw files will be preserved without modification, and processed datasets will be generated through reproducible scripts.

Primary Sources

CatPred-DB

BRENDA

SABIO-RK

UniProt

Variables

Enzyme sequence

EC class

Organism source

Substrate name / SMILES

Experimental kinetic values

Planned Data Handling

Remove duplicates

Missing value filtering

Log transformation

Outlier inspection

Standardization

Train / validation / test split

7. Feature Engineering Strategy

Enzyme Features

Sequence length

Amino acid composition

Hydrophobicity index

Charge ratio

Molecular weight estimate

Substrate Features

Molecular weight

Atom count

Ring count

Topological descriptors

Encoded categories

Embedding Features (Advanced)

Protein embeddings from pretrained models

Molecular fingerprints

8. Evaluation Design

Metrics

$$RMSE = \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2}$$

$$MAE \frac{1}{n} = \sum |y_i - \hat{y}_i|$$

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

Validation Strategy

Hold-out train/test split

5-fold cross validation

Robustness check across enzyme classes

Comparative Baselines

Mean predictor

Linear model

Tree ensemble model

Embedding-enhanced model

9. Scientific Visualization Plan

Planned figures:

1. Workflow diagram
2. Data distribution of target values
3. Predicted vs Actual scatter plot
4. Residual analysis plot
5. Model comparison bar chart
6. Feature importance ranking
7. SHAP explanation plot (optional)

10. Innovation / Bonus Directions

Possible high-score extensions:

- Compare raw target vs log-transformed target
- Multi-target joint prediction
- Sequence-only vs sequence+substrate comparison
- Lightweight stacking ensemble model
- Independent external validation subset
- SHAP biological interpretability study

11. Risk Management

Potential Risks

- Dataset incompleteness
- Limited computing resources
- High-dimensional feature sparsity

Solutions

Use reduced benchmark subset first

Prioritize baseline models

Use cloud notebook if necessary

Keep reproducible scripts

12. Timeline

Before April 30

Finish plan

Download data

Build repository structure

May 1 – May 15

Baseline model implementation

Initial experiments

May 16 – May 20

Data preprocessing refinement

Visualization

May 21 – May 31

Final report writing

Presentation preparation

13. Expected Contributions

This project is expected to provide:

1. A reproducible statistical learning benchmark for enzyme kinetics
2. Interpretable comparison across methods
3. Practical insights into biochemical feature relevance
4. A scalable path toward multimodal AI4Science modeling

References

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